

DEBUG LOG

Debug Log — Trek Ready App

A chronological record of bugs caught, their root causes, and how they were fixed.

For the engineering decisions behind each fix, see CLAUDE.md section 6.

Navigation & State Bugs

Inconsistent query param API (Phase 5)

- **Symptom:** Page nav and refresh broke when switching between Overview and Guidance.
- **Root cause:** Mixed `st.experimental_get_query_params`` and `st.query_params`` across the codebase.
- **Fix:** Standardised everything on `st.query_params``; added a normalisation helper.

Risk driver guidance link mismatch (Phase 5)

- **Symptom:** Clicking a high-risk driver link didn't reliably navigate to the focused section.
- **Root cause:** ``+`` encoding in URLs; focus values decoded inconsistently.
- **Fix:** Switched to percent-encoding with `quote(..., safe=")``; added `normalize_focus()``.

Sidebar section selector stuck (Phase 5)

- **Symptom:** Left nav could get stuck in Guidance; couldn't toggle back to Overview.
- **Root cause:** Streamlit widget state conflicts and improper session-state overrides.
- **Fix:** Simplified sidebar selection logic; removed conflicting state sync.

Page URL out of sync on refresh (Phase 5)

- **Symptom:** Refreshing sometimes returned to a previous page state.
- **Root cause:** Query params not updated consistently on section change.
- **Fix:** Added explicit query param sync; helper keeps browser URL aligned.

Tests passed but navigation bug remained (Phase 5)

- **Symptom:** All tests green; nav bug survived.
 - **Root cause:** Tests only covered helper output, not section resolution logic.
 - **Fix:** Expanded tests to cover query utilities, page resolution, link encoding, and guidance rendering.
-

Scoring Model Bugs

Fitness risk scored against total distance, not daily (Phase 6b)

- **Symptom:** Beginner doing 80 km over 10 days scored same fitness risk as 80 km in 3 days.
- **Root cause:** ``score_fitness_risk`` used raw route totals; ``trip_profile`` was never passed in.
- **Fix:** Added ``trip_profile`` param; compute ``moving_days = total_days - rest_days``; score daily load.

Hotel treated as higher logistics risk than backcountry hut (Phase 6b)

- **Symptom:** Hotel got +3 more logistics risk than a backcountry hut.
- **Root cause:** ``else`` branch caught both ``"Hotel"`` and ``"Campsite"`` and added 4 to both.
- **Fix:** Added explicit ``"Hotel"`` branch (+0) at the top; full ordered tier now explicit.

Risk label diverged from readiness score (Phase 6b)

- **Symptom:** UI showed "72/100 readiness" alongside "High" risk simultaneously.
- **Root cause:** ``risk_engine.py`` had its own independent scoring logic separate from ``scoring_model.py``.
- **Fix:** Rewrote ``estimate_risk`` to call the shared pipeline; risk label derived from readiness score.

Guided trips got no meaningful risk reduction (Phase 6b)

- **Symptom:** Beginner guided vs. self-guided differed only by 2 logistics points.
- **Root cause:** Guided/self-guided only touched ``score_logistics_risk``.
- **Fix:** Added ``apply_guided_adjustments()`` — reduces Fitness and Gear risk for guided trips.

Gear risk lodging tier inversion (same bug as logistics, found later) (Phase 6b)

- **Symptom:** Hotel had higher gear risk than backcountry hut.
- **Root cause:** Same ``else`` → +4 catch-all that was fixed in logistics existed in gear risk.
- **Fix:** Applied same explicit tier ordering to ``score_gear_risk``.

Altitude risk zero-band too high (Phase 6b)

- **Symptom:** W Trek (1200m max) scored 0 altitude risk despite mountain conditions.
- **Root cause:** Zero-risk threshold was ``< 1200m``.
- **Fix:** Lowered to ``< 800m``; added 800–1500m band (+3); rescaled upper bands.

Weather scoring hemisphere-blind (Phase 6b)

- **Symptom:** October Patagonia trip (southern summer) penalised as "Fall."
- **Root cause:** Season inferred from month name substrings — northern-hemisphere assumption baked in.
- **Fix:** Dropped season parsing entirely; score from ``route["climate"]`` field instead.

Gear risk had no feedback from the packing plan (Phase 6b)

- **Symptom:** Well-prepared hiker scored same gear risk as poorly-prepared one.
 - **Root cause:** `score_gear_risk` computed before the packing plan existed.
 - **Fix:** Two-pass flow: initial scores → build packing plan → `adjust_gear_risk_for_warnings()` → adjusted scores.
-

API & Environment Bugs

AI layer silently swallowed all errors (Phase 7d)

- **Symptom:** AI narratives showed "unavailable" with no explanation; no way to debug.
- **Root cause:** All three generation functions had bare `except Exception: return None`.
- **Fix:** Extracted shared `_call()` helper that logs errors to stderr before returning None.

`ANTHROPIC_API_KEY` conflicting with Claude Code auth (Phase 7d)

- **Symptom:** App used Claude Code's session key instead of the project key.
- **Root cause:** `load_dotenv()` default `override=False` deferred to whatever was already in the environment.
- **Fix:** Changed to `load_dotenv(override=True)` so `.env` file always wins.

Stale `None` cached in Streamlit on first AI failure (Phase 7d)

- **Symptom:** After fixing the API key, AI sections still showed "unavailable."
 - **Root cause:** `@st.cache_data` cached the `None` result permanently for the session.
 - **Fix:** Added `ttl=3600` so failed results expire after one hour.
-

Discovery & Map Bugs

OSM name search returns wrong route (Phase 8b)

- **Symptom:** Searching "Appalachian Trail" added "International Appalachian Trail, Québec."
- **Root cause:** Global Overpass name search returns first regex match; OSM name took priority over Wikipedia title.
- **Fix:** Swapped priority to prefer Wikipedia title (curated, English); OSM name only when no Wikipedia title.

Overpass global name search timing out (Phase 8b/map)

- **Symptom:** `HTTP Error 504: Gateway Timeout` when fetching trail geometry by name.
- **Root cause:** Global regex search over all OSM data is expensive; busy servers time out.
- **Fix:** Switched to bbox search ($\pm 1.5^\circ$ around route centre); name-match scoring picks best result from local results.

Elevation chart base altitude wrong for high-gain routes (Phase 8a/map)

- **Symptom:** Tour du Mont Blanc chart showed 0m start and 0m end; Chamonix is at 1035m.
 - **Root cause:** `base_elev = max(0, max_elev - total_gain)` → 0 when `total_gain > max_elev`.
 - **Fix:** Added `base_elevation_m` field to route library; chart uses that when present.
-

Python Compatibility Bugs

`str | None` type hints fail on Python 3.9 (Phase 8a)

- **Symptom:** `TypeError: unsupported operand type(s) for |: 'type' and 'NoneType'` at startup.
 - **Root cause:** Union type syntax `X | None` requires Python 3.10+ at runtime unless `from __future__ import annotations` is present.
 - **Fix:** Added `from __future__ import annotations` to `app.py` and `charts.py`.
-

Data / Logic Bugs

`max_value=21` on trip days broke Camino Francés (30-day route)

- **Symptom:** `StreamlitValueAboveMaxError` on selecting Camino Francés.
- **Root cause:** Hard-coded `max_value=21` predated multi-week routes.
- **Fix:** Raised `max_value` to 60 days; `rest_days` max raised from 7 to 14.